

Yizhou Lu

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EDUCATION

University of Southern California

Master of Science in *Computer Science*

Los Angeles, CA

Aug 2025 – May 2027

University of Wisconsin–Madison

Bachelor of Science in *Computer Sciences* | Bachelor of Science in *Mathematics*

Madison, WI

Sep 2022 – May 2025

INDUSTRIAL EXPERIENCE

Software Engineer Intern | **TalkMeUp** | Los Angeles, CA

Jun 2026 – Aug 2026

- Develop FreshLens, a computer-vision produce-freshness platform using FastAPI, YOLOv8, and Claude API to detect produce items, score freshness (0–100), and generate automated discount worklists and rescue recipes.
- Implement multi-item object detection with YOLOv8, using produce-class filtering and confidence thresholding to replace whole-image fallbacks (e.g., improved precision from 9 raw detections to 4 accurately labeled items, eliminating false positives).
- Build a Next.js/React (TypeScript, Tailwind) frontend module to render live detection feeds with dynamically scaled bounding-box overlays and real-time statistics for stakeholder demos.
- Engineered features within an 8-person team by establishing automated CI/CD guardrails (GitHub Actions, pytest, Ruff, secret scanning) and enforcing strict modular isolation.

Software Engineer Intern | **Baowin Steel LLC** | Houston, TX

May 2025 – Aug 2025

- Built a full-stack e-commerce trading platform using Django, React, and MySQL, supporting authentication, listings, and transactions which work as the foundation for the company's future platform.
- Designed and implemented RESTful APIs for account management and listing workflows.
- Deployed the application on AWS (EC2, RDS) and set up CI/CD pipelines, migrating from on-prem infrastructure.

Software Engineer Intern | **Lark Technologies Pte. Ltd.** | China

May 2024 – Jul 2024

- Developed an AI bot to automate Google Sheets workflows using **Python** and **JavaScript**, reducing manual data handling.
- Customized APIs and fine-tuned **LLM-based workflows** on Coze.com for structured data manipulation.
- Optimized performance with **multithreading and batch processing**, automating a 2,000-row customer-potential analysis and replacing ~2 weeks of manual work with ~4.5 hours of unattended processing.
- Integrated systems using **RESTful APIs and OAuth**, collaborating with cross-functional teams to define technical requirements.

PROJECTS

[JobSignal](#) – AI Job-Search Platform

Jul 2026 – Present

- Built an end-to-end RAG matching pipeline that extracts resume profiles with Claude, retrieves candidates from 8,400+ jobs using pgvector HNSW search, and reranks them with LLM scoring and per-job rationales, rendered as an interactive 3D galaxy at roughly \$1–2 per user per month in LLM cost through per-user quotas.
- Grew the job pool 250× from 34 to 8,400+ listings, 96% of them confirmed H1B sponsors, by building a pluggable multi-source ingestion system that crawls Greenhouse and Lever boards seeded from 27,428 real DOL filing records and refreshes on autonomous daily crawls.
- Engineered on Vercel with 254 unit tests and Playwright E2E coverage, row-level-security multi-tenancy, and prompt-injection hardening, developed through a 60+ PR workflow of issues and merge trains driven by parallel AI coding agents..

[GoThin](#) – AI Fitness Coach

Jun 2026 – Present

- Developed a full-stack AI fitness platform (Next.js, TypeScript, Supabase) using Claude API to transform food photos and natural-language logs into structured nutrition metrics, incorporating confidence-scored macro estimates and linear-regression date forecasting.
- Engineered LLM cost efficiency with model routing, **prompt caching (~7× cheaper input tokens)**, schema-constrained structured outputs, and a trace dashboard tracking per-call cost, latency, and cache-hit rate.

Twitch Gadget – Content-Based Recommendation System

Sep 2025 – Nov 2025

- Built a full-stack platform using Spring Boot, React, and MySQL, deployed on AWS.
- Integrated Twitch API to retrieve live streams, clips, and replays.
- Implemented content-based recommendations using user favorites and game metadata.
- Secured authentication with Spring Security and improved performance using Caffeine caching.

SKILLS

- Programming Languages: Python, Java, C++, JavaScript/TypeScript, C#
- Web & Backend: React, Next.js, Node.js, Django, FastAPI, Spring Boot
- Databases & Cloud: MySQL, PostgreSQL (Supabase), AWS (EC2, RDS), Docker, CI/CD (GitHub Actions, GitLab)
- AI & CV: LLM APIs (Claude, OpenAI), LangChain, prompt caching & structured outputs, YOLO/Ultralytics, embedding API
- Practices: RESTful APIs, Object-Oriented Design, Agile Development, RAG
- AI-assisted Development: Claude Code, Codex, Kimi Code, Customized AI Agent Harness based on spec-driven-development